

Bowen Song

1400 S Figueroa St – Los Angeles, CA 90015 – USA

☎ +1 (857) 318-5369 • ✉ bowenson@usc.edu • 🌐 bowenlandsong.github.io
in songbowen • 🔄 Bowenlandsong • 🆔 0000-0002-5071-3880



Summary

CS PhD at USC specializing in **multimodal representation learning, conditional computation, and scalable federated AI systems for edge deployment**. My research focuses on learning shared latent embeddings across heterogeneous time-synchronized modalities (wearables, motion capture, force sensing) for cross-modal prediction and personalization under strict latency and resource constraints. I design gated and mixture-based architectures to balance global generalization with task- and subject-specific specialization, including deployment in on-device settings. Industry experience at **eBay (Core Ranking)** developing large-scale Transformer-based Mixture-of-Experts systems for multi-objective modeling. Publications in *IEEE TBME* and *IEEE CLOUD*.

Education

University of Southern California

Ph.D. in Computer Science

Los Angeles, CA

2020–2026 (expected)

Research: Multimodal representation learning for time-series sensor data; low-rank embedding learning; conditional computation and mixture-based architectures; federated personalization under privacy and edge device (phone) constraints; scalable ML systems. Publications: *IEEE TBME* (2025), *IEEE CLOUD* (2022). Advisor: Leana Golubchik.

Boston University

M.S. in Electrical & Computer Engineering

Boston, MA

2019

Publications: *IEEE Allerton* (2019). Advisor: Ari Trachtenberg

University of Victoria

B.Eng. in Electrical Engineering. Focus: signal processing and optimization

Victoria, BC

2017

Experience: Multimodal AI & Large-Scale Modeling

eBay Inc.

PhD Research Intern, Core Ranking

San Jose, CA

May–Aug 2024

- Designed and deployed **Transformer-based Mixture-of-Experts (MoE)** architectures integrating language and structured signals for large-scale ranking, enabling conditional computation under multi-objective optimization (conversion, engagement, long-term value).
- Developed adaptive expert routing and capacity allocation mechanisms for dynamically evolving traffic distributions, improving specialization while preserving global generalization.
- Built scalable distributed training pipelines on enterprise **Krylov** across multi-GPU clusters; implemented robust monitoring, evaluation, and experimentation frameworks to support reliable large-scale model iteration.

University of Southern California

Graduate Researcher, Personalized AI for Health (QEDLAB)

Los Angeles, CA

Aug 2020–Present

- Designed a **federated multi-modal multi-task representation learning framework** for time-aligned biomechanical signals, learning shared latent representations across heterogeneous sensing platforms (wearables, motion capture systems, instrumented force plates).
- Designed **cross-modal generative and predictive models** operating in shared embedding space (e.g., wearable sensors → lab-grade instrumented surface), enabling scalable multi-output inference for injury risk estimation, counterfactual (what-if) modeling, and live assistive control (e.g., exoskeletons) under strict latency and resource constraints.
- Proposed **conditional expert routing and gating mechanisms** to balance global generalization with subject- and task-specific specialization.
- Investigated robustness under distribution shift (unseen subjects, activities, environments) using mixture-based and federated optimization strategies.
- Built end-to-end training and evaluation pipelines for large multimodal time-series datasets; benchmarked linear, recurrent, and deep baselines including transformers, attention-based fusion, and diffusion-based generative models.

Experience: Cloud & Distributed Systems

eBay Inc.

PhD Research Intern, Infrastructure

San Jose, CA

May–Aug 2023

- Owned **Kafka Connector** integration into a multi-datacenter **ClickHouse** store with real-time ingestion; ensured **exactly-once** semantics and streamlined upgrades across environments.

Red Hat (IBM)

Boston, MA

Software Engineer, *OpenShift*

Jun 2019–Aug 2020

- Owned and advanced the **Operator Lifecycle Manager (OLM)**, a core Kubernetes subsystem powering operator installation and lifecycle management across OpenShift; improved dependency resolution, upgrade safety, and version orchestration for production clusters.
- Led cross-team integration of **Helm** into the Operator Framework, enabling bidirectional packaging of operators and significantly expanding ecosystem interoperability; contributed enhancements to the upstream **Kubernetes scheduler and descheduler**.
- Designed and deployed a **statistical flake-detection system** integrated into CI pipelines, reducing false-positive failures and accelerating root-cause analysis across large-scale distributed builds.

USC

Los Angeles, CA

Researcher, *Cloud Resource Management (QEDLAB)*

2020–2022

- Developed a **queueing-theoretic framework for hybrid-cloud federations**, proving that incremental resource sharing monotonically improves system utility, with full federation achieving optimal performance under QoS constraints. Designed an incentive-compatible revenue allocation mechanism that fairly distributes federation profits and discourages free-riding in a fully decentralized setting (*IEEE CLOUD 2022*).

Selected Publications

- B. Song *et al.*, “Estimating Ground Reaction Forces from Inertial Sensors,” *IEEE Trans. Biomedical Engineering*, 2024.
- B. Song, M. Paolieri, L. Golubchik, “Performance and Revenue Analysis of Hybrid Cloud Federations with QoS Requirements,” *IEEE CLOUD*, 2022.
- B. Song, A. Trachtenberg, “Scalable String Reconciliation by Recursive Content-Dependent Shingling,” *Allerton*, 2019.
- R. Liu, B. Song *et al.*, “Contextual Mixture of Ranking Models,” *under review*, 2025.
- B. Song *et al.*, “Federated Multi-Task Learning for Biomechanical Signals from Wearable Sensors,” *under preparation*, 2026.

Skills

ML: PyTorch, TensorFlow, CUDA; multimodal modeling, Transformer architectures, mixture-of-experts, conditional computation, time-series representation learning

Data/Infra: Kubernetes, Containers, **Kafka**, **ClickHouse**, Ray, Kubeflow; metric/experiment platforms; GPU clusters

Languages: Python, Go, C/C++, Java, Scala, MATLAB, R; $\text{\LaTeX}$

Selected Open Source Projects

Operator Lifecycle Manager: Core contributor to the Operator Framework, a foundational Kubernetes subsystem for operator lifecycle management; advanced dependency resolution, upgrade safety, and ecosystem interoperability across production OpenShift clusters.

Flake Analyzer: Designed and deployed a statistical CI reliability system that detects and classifies flaky tests at scale, reducing false-positive failures and accelerating distributed system debugging workflows.

RCDS Protocol: Developed the Recursive Content-Dependent Shingling (RCDS) protocol for bandwidth-efficient distributed file synchronization; achieves superior performance to `rsync` under high-frequency large-file updates.

CoT-Rec: Ranking-guided Chain-of-Thought framework for improving LLM reasoning via candidate reranking; enables systematic evaluation of reasoning quality and selection strategies.

Service & Community Involvement

Reviewer for Journals and Conferences: ACM SIGMETRICS, IEEE CLOUD, *Sensors*.

Conference: Co-hosted Red Hat’s 2020 Summit (20k+ attendees). Presenter at IEEE CLOUD 2022; DevConf.US 2020.